Software Test Plan (STP)

Electronics Store Inventory Management System

Project	Electronics Store Inventory Management System
Version	1.0
Team	H9
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Technology Stack	Python (Django)
Date	14-10-2026
Status	Draft / For Review

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Revision History

Version	Date	Author	Change Summary
1.0	14-10-2026	Team H9	Complete test plan based on SRS v1.0, the Phase 1 test plan extract and the sprint plan

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1. Introduction

Purpose	This document defines the test plan for the Electronics Store Inventory Management System v1.0. It describes what will be tested, how, when, by whom, and when testing is complete.
Scope	Testing covers login and logout, role-based access, adding, viewing, searching, updating and deleting components, availability and low-stock identification, dashboard, reports, input validation, error handling, change tracking, data storage, performance, reliability, usability, browser compatibility, backup / restore and security.
References	SRS v1.0 (01_COMPLETE_SRS_IEEE_STYLE), Architecture & Design Specification v1.0, Test Cases (03_COMPLETE_TEST_CASES_15), GitHub Project "SE Minproject" (backlog and sprints), OWASP Top 10.
Definitions	SRS – Software Requirements Specification; RTM – Requirements Traceability Matrix; TC – Test Case; UAT – User Acceptance Testing; CSRF – Cross-Site Request Forgery; ESIM-F / NF / SR – Functional / Non-Functional / Security requirement IDs.

2. Test Items

Test item (module)	What it does
Accounts – AuthModule, SessionManager	Login and logout
Accounts – AuthorizationModule	Roles and permissions
Inventory – InventoryModule	Add, update, delete / deactivate components
Inventory – SearchFilterModule	Search and filter
Inventory – ValidationModule	Input checks
Inventory – AvailabilityService, StockAlertService	Availability and low-stock items
Reports – DashboardModule, ReportModule	Dashboard totals and reports (CSV)
Audit – AuditLogModule	Change history
Core – ErrorHandler, database layer	Error pages, persistent storage, backup / restore

3. Features to be Tested

Features are mapped to SRS requirement IDs and to the sprint in which they are built (from the GitHub sprint plan).

Req ID	Feature	Test case	Sprint
ESIM-F-001	User login	TC-AUTH-01	Sprint 1
ESIM-F-002	Logout and session end	TC-AUTH-02	Sprint 1
ESIM-F-003	Add component	TC-INV-01	Sprint 1
ESIM-F-004	View inventory	TC-INV-02	Sprint 1
ESIM-F-005	Search and filter	TC-SEARCH-01	Sprint 1
ESIM-F-006	Update component	TC-INV-03	Sprint 1
ESIM-F-007	Delete / deactivate component	TC-INV-04	Sprint 1
ESIM-F-008	Availability status	TC-AVAIL-01	Sprint 1

Req ID	Feature	Test case	Sprint
ESIM-F-009	Low-stock identification	TC-LOW-01	Sprint 1
ESIM-F-010	Dashboard	TC-DASH-01	Sprint 2
ESIM-F-011	Inventory reports	TC-REP-01	Sprint 2
ESIM-F-012	Input validation	TC-VAL-01	Sprint 1
ESIM-F-013	Persistent storage	TC-DATA-01	Sprint 1
ESIM-F-014	Role-based authorization	TC-SEC-01	Sprint 1
ESIM-F-015	Error handling	TC-ERR-01	Sprint 2
ESIM-F-016	Change tracking	TC-AUD-01	Sprint 2
ESIM-NF-001	Response time ≤ 2 s for 90% of requests	TC-PERF-01	Sprint 2
ESIM-NF-002	Works with 1000+ component records	TC-PERF-02	Sprint 2
ESIM-NF-003	Transaction reliability	TC-REL-01	Sprint 2
ESIM-NF-004	99% availability in test run	TC-REL-02	Sprint 2
ESIM-NF-005	Usability for new users	TC-UX-01	Sprint 2
ESIM-NF-006	Modular code structure	Code review	Sprint 2
ESIM-NF-007	Chrome, Edge, Firefox compatibility	TC-COMP-01	Sprint 2
ESIM-NF-008	Backup and restore	TC-REC-01	Sprint 2
ESIM-SR-001..006	Security requirements (see 5.1)	TC-SEC-01..06	Sprint 2

4. Features Not to be Tested

- Online purchasing, payment processing, supplier payments, accounting and demand forecasting – outside the project scope (SRS 1.2).
- Internal behaviour of the Django framework and the database engine – assumed tested by their maintainers.
- Production-scale load above 1,000 records; TC-PERF-02 at 1,000 records remains in scope.
- Penetration testing of any system other than the team's own development / test environment.

5. Test Approach / Strategy

Test levels

Level	What is tested	How
Unit	Single functions and models: availability rule, validation rules, low-stock query	Django TestCase / unittest
Integration	Views + services + database working together (e.g. add component writes a ChangeLog)	Django test client
System	Complete user flows in a browser: login → add → search → dashboard → report	Manual tests + Selenium
Acceptance (UAT)	Acceptance criteria of each user story on the GitHub board	Demo at end of each sprint

Test types

- **Functional** – each FR against its acceptance criteria and test case.

- **Boundary value** – quantity below, equal to and above the minimum stock level; zero and negative values.
- **Regression** – rerun earlier tests after every merged pull request (automated with GitHub Actions).
- **Performance** – response time and 1000+ records (Locust or JMeter).
- **Usability and compatibility** – new-user task test; Chrome, Edge and Firefox.
- **Security** – see Section 5.1.

Entry and exit criteria

Entry criteria	Feature merged to the main branch, test data loaded, test environment running, test cases written.
Exit criteria	100% of planned test cases executed; all High-priority tests passed; 0 critical or high defects open; RTM updated with results; acceptance criteria of sprint stories met.

5.1 Security Validation

Security test	Requirement	Expected result	Test case
Open protected pages without login	ESIM-SR-001	Redirected to login; no data shown	TC-SEC-01
Staff / Manager try admin actions, incl. direct POST requests	ESIM-SR-002	Blocked with 403; no data changed	TC-SEC-02
Inspect stored passwords	ESIM-SR-003	Only hashed values (Django PBKDF2)	TC-SEC-03
Send form request without CSRF token	ESIM-SR-004	Request rejected (403)	TC-SEC-04
Enter SQL-injection and script text in forms	ESIM-SR-005	Input rejected or safely escaped; records unchanged	TC-SEC-05
Check logs after login, logout and failed login	ESIM-SR-006	No passwords or secret keys in logs	TC-SEC-06
Session test after logout (Back button)	ESIM-F-002	Re-login required	TC-AUTH-02

An automated scan with OWASP ZAP will be run on the local test server only.

6. Test Environment

Item	Details
Hardware	Team members' laptops (Windows / macOS / Linux), 8 GB RAM or more
Software	Python 3, Django, SQLite (test database), application v1.0 from the GitHub main branch
Browsers	Latest Chrome, Edge and Firefox
Tools	Django test runner, Selenium, Locust / JMeter, OWASP ZAP, GitHub Actions (CI), GitHub Issues (defect tracking)
Test data	Sample users for each role (Admin, Staff, Manager); 20 sample components for functional tests; 1000+ generated components for performance tests

7. Test Schedule

Milestone	Dates
Test plan and test case design	12-Oct-2026 to 16-Oct-2026 (Sprint 0 – dry run)
Test environment and test data setup	16-Oct-2026 to 18-Oct-2026
Sprint 1 testing – core features (FR-01..09, FR-12..14)	19-Oct-2026 to 23-Oct-2026

Milestone	Dates
Sprint 1 review and demo video	23-Oct-2026
Sprint 2 testing – dashboard, reports, NFR and security tests	26-Oct-2026 to 30-Oct-2026
Regression and UAT	29-Oct-2026 to 30-Oct-2026
Test summary report	30-Oct-2026

8. Test Deliverables

- Software Test Plan (this document)
- Test cases (03_COMPLETE_TEST_CASES_15, updated with results)
- Automated test scripts (Django tests, Selenium, Locust) in the repository
- Test data scripts
- Test execution logs (GitHub Actions runs)
- Defect reports (GitHub Issues with label "bug")
- Updated RTM and Test Summary Report

9. Roles and Responsibilities

Testing responsibilities follow the story assignments on the GitHub board – each member tests the stories they build.

Role	Name	Responsibility
QA Lead / Security Tester	Swara Potdar	Maintain this plan, coordinate test runs; test login, roles and all security tests (SR-01..06)
Test Engineer	Simret M Kotian	Test add / update, logout, dashboard; performance and scalability tests (NFR-01, 02); code review (NFR-06)
Test Engineer	Shriya S R	Test view, search, delete, reports, change tracking; reliability and browser tests (NFR-03, 07)
Test Engineer	Ahana S S	Test availability, low stock, validation, storage, error handling; uptime, usability and backup tests (NFR-04, 05, 08)
Repo Owner / Reviewer	Simret M Kotian	Review and merge pull requests after tests pass
Approver	Prof. Ambika K (Course Coordinator)	Review results and sign off

10. Risks and Mitigation

Risk	Mitigation
Features finished late in the sprint, leaving little test time	Write test cases before coding; test each pull request as it is raised
Test environment differs between laptops	Use requirements.txt and the same SQLite test database; run tests in GitHub Actions
Not enough realistic data for performance tests	Script to generate 1000+ components

Risk	Mitigation
Security tests missed due to time	Security tests planned as separate stories with story points in Sprint 2

11. Assumptions & Dependencies

- Features are merged to the main branch before they are tested.
- Each team member has Python, Django and the browsers installed.
- GitHub Actions is enabled for the repository (as required in Sprint 1).
- Requirements in SRS v1.0 do not change during a sprint; changes are agreed at sprint planning.

12. Suspension & Resumption Criteria

Suspend testing if	The build does not start, login does not work, or more than 30% of planned test cases are blocked; or the test database is corrupted.
Resume testing when	The blocking defect is fixed and merged, the environment is working again, and a quick smoke test (login + view inventory) passes.

13. Test Case Management & Traceability

Test cases are stored in the repository and linked to GitHub issues (stories). The RTM in the SRS (Section 8) maps each requirement to its test case; the Status column is updated after each sprint (N = Not tested, P = In progress, A = Passed).

Requirement	GitHub story	Test case(s)	Status
ESIM-F-001 Login	FR-01 (#1)	TC-AUTH-01	N
ESIM-F-002 Logout	FR-02 (#2)	TC-AUTH-02	N
ESIM-F-003 Add component	FR-03 (#3)	TC-INV-01	N
ESIM-F-008 Availability	FR-08 (#10)	TC-AVAIL-01	N
ESIM-F-012 Validation	FR-12 (#14)	TC-VAL-01	N
ESIM-F-014 Authorization	FR-14 (#16)	TC-SEC-01	N
ESIM-NF-001 Performance	NFR-01 (#19)	TC-PERF-01	N
ESIM-SR-004 CSRF	SR-04 (#30)	TC-SEC-04	N

(Sample rows shown; the full mapping for all 30 requirements is in Section 3 and in the SRS RTM.)

14. Test Metrics & Reporting

Metrics collected

- % of test cases executed and % passed / failed
- Number of defects by severity, and defects still open
- Requirement coverage (requirements with at least one passed test)
- Story points completed per sprint (from the GitHub board)

Reports

- Status update at the end of each sprint (GitHub board + Insights)

- Final Test Summary Report at the end of Sprint 2

15. Approvals

Role	Name	Signature / Date
QA Lead	Swara Potdar	
Repo Owner / Dev Lead	Simret M Kotian	
Course Coordinator	Prof. Ambika K	

Appendix A — Minimum Representative Test Case Set

The plan maps the full 29-case catalog in the separate Test Cases deliverable. This appendix includes ten representative cases in the plan itself, covering functional, non-functional, and security requirements as required by the Phase 1 checklist. All remain planned/not tested in the Phase 1 baseline.

Case / requirement	Procedure	Expected result / pass criterion
TC-AUTH-01 ESIM-F-001	Submit valid credentials, then invalid credentials.	Valid login creates an authorized session; invalid login does not.
TC-INV-01 ESIM-F-003	As an authorized administrator, submit one valid component record.	Exactly one record is created with the submitted values.
TC-SEARCH-01 ESIM-F-005	Search by component ID/name and filter by category.	Every result matches the selected query/filter; no unrelated result appears.
TC-AVAIL-01 ESIM-F-008	Check quantities 0, below threshold, equal to threshold, and above threshold.	Availability is Out of Stock, Low Stock, Low Stock, and In Stock respectively.
TC-VAL-01 ESIM-F-012	Submit missing fields, duplicate ID, negative quantity, and invalid price.	Each invalid submission is rejected; existing valid data remains unchanged.
TC-DATA-01 ESIM-F-013	Create record; log out, log in, and retrieve it.	All saved fields remain unchanged and retrievable.
TC-DASH-01 ESIM-F-010	Load a fixture with active, inactive, and low-stock records; compare metrics.	Active count, available units, and low-stock count equal independent fixture calculations.
TC-PERF-01 ESIM-NF-001	Time 10 list, 10 search, 10 add, 10 edit, and 10 deactivate requests.	At least 45 of 50 requests finish in <2.0 seconds.
TC-REL-02 ESIM-NF-004	Probe the approved endpoint every 60 seconds for the scheduled 12-hour window.	Availability report is ≥99.00%; samples and report are retained. Do not infer a pass before the window ends.
TC-SEC-01 ESIM-SR-001 / ESIM-F-014	Request a protected page while logged out, then attempt a restricted operation as a user without permission.	Protected content is not disclosed; restricted write is denied and database state is unchanged.

Execution evidence shall record actual result, date, build/commit, environment, tester, and linked defect. TC-REL-02 and TC-UX-01 require their planned manual/operational evidence; automated test-client checks alone do not satisfy them.

Standards basis: The plan follows the supplied STP template and a numbered, traceable test-plan structure informed by IEEE Std 829 and the ISO/IEC/IEEE 29119 series. It does not claim conformance certification.